

hospital

Req2Design · IEEE 830-1998 aligned

Document Control

Version	Date	Author	Change Summary
1.0	2026-04-09	abdulhaq	Initial generated draft for structured review.

Document ID: SRS-20260409-200014

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1: Introduction

1.1: Purpose

Purpose: To provide a comprehensive medical appointment scheduling system for patients and healthcare providers.

1.2: Scope

Scope: This document outlines the requirements for the hospital appointment management system focusing on functionalities related to patient-doctor interactions, including finding doctors, booking appointments, managing schedules, handling walk-ins, and maintaining records.

1.3: Definitions/Acronyms

- **DOCTOR:** Healthcare professional providing medical services.
- **PATIENT:** Individual seeking medical care.
- **WALK-IN:** Patient arriving without pre-scheduled appointment.

1.4: References

References: IEEE 830-1998

1.5: Overview

Overview: This system will facilitate efficient appointment management through intuitive interfaces for both patients and doctors, ensuring timely access to healthcare while minimizing administrative overhead.

2: Overall Description

2.1: Product Perspective

Product Perspective: The system is designed to streamline the process of making and managing medical appointments between patients and doctors within a hospital setting.

2.2: Product Functions

- Patients can search for doctors based on specialties and book appointments.
- Patients receive reminders via SMS or email about upcoming appointments.
- Doctors can manage their schedules, block unavailable time slots, and record visit outcomes.
- Walk-in patients can be accommodated and scheduled accordingly.

2.3: User Characteristics

User Characteristics: Users include patients who require medical attention and doctors responsible for treating them.

2.4: Constraints

Constraints: The system must not allow double bookings, ensure that all booked times are unique, and support basic patient history visibility during booking.

2.5: Assumptions/Dependencies

Assumptions/Dependencies: The system assumes reliable internet connectivity and standard SMS/email delivery mechanisms.

3: Specific Requirements

3.1: External Interface Requirements

- **User Interface:** Graphical interface for patients and doctors.
- **Hardware Interface:** Standard computer systems and mobile devices.
- **Software Interface:** Integration with existing hospital databases.
- **Communication Interface:** SMS gateway for sending reminders and email server for notifications.

3.2: Functional Requirements

- **FR-01:** Find Doctor By Specialty

Input: Patient searches for doctor by specialty name.

Processing: System retrieves list of available doctors matching the specialty.

Output: List of doctors displayed to the patient.

Priority: High

- **FR-02:** Book Appointment

Input: Patient selects preferred date, time slot, and doctor from the list.

Processing: System checks for conflicts and reserves the selected slot if available.

Output: Confirmation message sent to the patient.

Priority: High

- **FR-03:** Get Reminders

Input: Patient provides contact information.

Processing: System sends reminder messages at specified intervals before the appointment.

Output: Reminder sent via SMS or email.

Priority: Medium

- **FR-04:** Block Unavailable Slots

Input: Doctor marks specific dates/times as unavailable.

Processing: System updates schedule to reflect these unavailability periods.

Output: Updated schedule shown to other users.

Priority: High

- **FR-05:** View Daily Schedule

Input: Doctor logs into the system.

Processing: System displays the full day's schedule including reserved and blocked slots.

Output: Detailed schedule presented to the doctor.

Priority: High

- **FR-06:** Mark Completed Visits

Input: Doctor indicates completion status after seeing a patient.

Processing: System records the outcome of the visit.

Output: Visit marked as complete in the database.

Priority: High

- **FR-07:** Handle Walk-In Patients

Input: Patient arrives unexpectedly without a reservation.

Processing: System allows the doctor to add a new entry for the walk-in.

Output: Walk-in appointment added to the schedule.

Priority: Medium

- **FR-08:** Reschedule Appointments

Input: Patient requests to change the original appointment details.

Processing: System finds alternative slots and rebooks the appointment.

Output: New confirmation message sent to the patient.

Priority: Medium

- **FR-09:** Cancel Appointments

Input: Patient cancels the previously made appointment.

Processing: System removes the reservation and makes the slot available again.

Output: Cancellation confirmed to the patient.

Priority: Medium

Non-Functional Requirements

- **Usability:** The system shall be easy to navigate and understand for both patients and doctors.
- **Reliability:** The system must accurately track all appointments and prevent double bookings.
- **Performance:** The system shall respond quickly to user inputs and handle up to 100 concurrent users efficiently.
- **Portability:** The system shall run on various operating systems and devices commonly used in hospitals.

4: System Features

Feature 1: Patient Registration

- Allows patients to create accounts and store personal information securely.
- Ensures that patient data is protected according to privacy laws.

Feature 2: Doctor Login

- Enables doctors to log in and access their schedule and patient history.
- Provides secure authentication methods to protect sensitive medical information.

Feature 3: Appointment Management

- Facilitates booking, rescheduling, canceling, and marking appointments as completed.
- Manages walk-in patients effectively and integrates seamlessly with existing hospital workflows.

Feature 4: Notification System

- Sends automated reminders to patients and doctors via SMS and email.
- Tracks the status of sent notifications and ensures high delivery rates.

Feature 5: Audit Trail

- Maintains a log of all significant events related to appointments, such as changes in schedules or cancellations.
- Supports auditing and compliance with regulatory standards.

End of Document.